River Araeth (Tributary of the River Tywi)

Macroinvertebrate and Water Quality Survey Report

Survey Date: 21 July 2026

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1. Introduction

A comprehensive macroinvertebrate and water quality survey was undertaken on the River Araeth, a tributary of the River Tywi, on 21 July 2026. The purpose of the survey was to assess the ecological condition of the river through biological monitoring using macroinvertebrates alongside chemical water quality analysis.

Macroinvertebrates are recognised as reliable indicators of long-term water quality because different groups exhibit varying tolerances to pollution. When combined with chemical analysis, they provide a comprehensive assessment of the ecological health of a river.

Sampling was undertaken using the standard freshwater biological monitoring method consisting of a **3-minute kick sample** followed by a **1-minute stone search** using a **1 mm mesh sampling net**.

2. Survey Conditions

River: River Araeth (Tributary of the River Tywi)

Date: 21 July 2026

Sampling Method

- Three-minute kick sample
- One-minute stone search
- 1 mm mesh sampling net

Air Temperature: 18.8 °C

Water Temperature: 14.4 °C

Survey conditions were favourable with good river flow and excellent habitat diversity. The water temperature of 14.4 °C is well suited to supporting pollution-sensitive aquatic invertebrates while maintaining high dissolved oxygen concentrations.

3. Water Chemistry Results

Parameter	Result
pH	7.4
Ammonia	0.20 mg/L
Nitrate	0.17 mg/L
Phosphate	0.09 mg/L

Water Chemistry Assessment

The chemical analysis indicates generally good water quality.

The pH of 7.4 is neutral and suitable for sustaining a healthy freshwater ecosystem. Ammonia, nitrate and phosphate concentrations were all low, suggesting minimal nutrient enrichment or organic pollution at the time of sampling.

Overall, the chemical results support the biological findings and indicate a healthy aquatic environment.

4. Macroinvertebrate Results

Taxon	Number Recorded
Goeridae	1
Leptoceridae (*Athripsodes aterrimus*)	8
Hydropsychidae	2
Leuctridae	3
Chloroperlidae (*Siphonoperla* spp.)	20
Ephemeridae	3
Heptageniidae (*Ecdyonurus* spp.)	4
Simuliidae	2
Gammaridae (*Gammarus* spp.)	14

Oligochaeta	1

Total organisms recorded: 58

5. BMWP Assessment

Family	BMWP Score
Goeridae	10
Leptoceridae	10
Hydropsychidae	5
Leuctridae	10
Chloroperlidae	8
Ephemeridae	10
Heptageniidae	10
Simuliidae	5
Gammaridae	6
Oligochaeta	1

BMWP Score: 75

Number of Scoring Families: 10

Average Score Per Taxon (ASPT): 7.5

Biological Assessment

The BMWP score of **75** and ASPT of **7.5** indicate **Good ecological quality**. The macroinvertebrate community was dominated by pollution-sensitive Ephemeroptera (mayflies), Plecoptera (stoneflies) and Trichoptera (caddisflies), collectively known as the **EPT group**, which are recognised as indicators of clean, well-oxygenated rivers.

The abundance of **Chloroperlidae (Siphonoperla spp.)**, together with representatives of **Goeridae**, **Leptoceridae**, **Leuctridae**, **Ephemeridae** and **Heptageniidae**, demonstrates that the River Araeth provides excellent habitat capable of supporting a diverse aquatic community.

The presence of only a single oligochaete worm and relatively few pollution-tolerant taxa indicates little evidence of organic pollution at the survey site.

6. Interpretation

Both the biological and chemical assessments demonstrate that the River Araeth is functioning as a healthy tributary of the River Tywi.

The diverse macroinvertebrate assemblage is typical of a well-oxygenated river with good habitat quality and limited environmental stress. The water chemistry supports these findings, with neutral pH and low nutrient concentrations indicating favourable conditions for aquatic life.

Although this survey represents a single sampling event, the results provide a valuable baseline for future monitoring and suggest that the river currently supports a healthy freshwater ecosystem.

7. Overall Assessment

The River Araeth supported a diverse macroinvertebrate community dominated by pollution-sensitive taxa. The BMWP score of **75** and ASPT of **7.5**, together with the chemical analysis, indicate that the site was in **Good ecological condition** at the time of sampling.

No evidence of significant organic pollution was detected during the survey. The combination of favourable physical conditions, healthy macroinvertebrate diversity and good chemical water quality suggests that the River Araeth is functioning as a healthy tributary of the River Tywi.

Survey Summary

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Sampling Method: 3-minute kick sample and 1-minute stone search using a 1 mm mesh net.

Air Temperature: 18.8 °C

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Water Chemistry

- pH: 7.4
- Ammonia: 0.20 mg/L
- Nitrate: 0.17 mg/L
- Phosphate: 0.09 mg/L

Total Macroinvertebrates Recorded: 58

BMWP Score: 75

ASPT: 7.5

Ecological Classification: Good

Conclusion

The River Araeth exhibited good ecological quality, supporting a diverse assemblage of pollution-sensitive macroinvertebrates indicative of a healthy, well-oxygenated freshwater environment. The chemical analysis corroborated the biological assessment, with all measured parameters falling within acceptable limits for a healthy upland tributary. Continued periodic monitoring is recommended to establish long-term trends and ensure the continued ecological integrity of this important tributary of the River Tywi.